Seat Majorities without Vote Majorities: Coalition Inversions in Brazil's Chamber of Deputies

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Abstract

Proportional representation is usually evaluated party by party, by comparing each party's share of seats with its share of votes. Yet fragmented legislatures are governed through combinations of parties, so small party-level deviations can become consequential when parties are combined. Brazil provides a useful setting because the Chamber of Deputies is assembled from twenty-seven separate district elections. Several institutional steps separate national votes from final party seat totals, and no national compensatory allocation reconciles them. I ask whether politically meaningful sets of parties can hold a legislative majority despite receiving less than half of all valid federal-deputy votes nationwide, a condition known as coalition inversion. Using the 2014, 2018, and 2022 elections, I follow the party sets represented in presidential cabinets and examine potential coalitions restricted by ideological proximity. Cabinet inversions appear following all three elections, as do inversions among minimal connected winning coalitions. In 2022, a seven-party bloc of ideologically adjacent parties combined 45.26 percent of the vote with 258 of 513 seats. Party-level deviations from proportionality can therefore have a distinct coalition-level consequence: they can determine which combinations of parties command a legislative majority.

1 Introduction

Electoral systems translate votes into seats. Under proportional representation, the quality of that translation is usually assessed by comparing each party's share of the vote with its share of the seats (Kaminski 2026). The comparison is indispensable, but it stops at the level of the units that contest the election. In fragmented legislatures, no single party need command a majority, and the majorities that organize government and legislative action are assembled from several parties.

Combining parties changes the significance of their individual deviations from proportionality. One party may receive slightly more seats than its national vote share implies, while another receives slightly fewer. When parties are grouped together, these deviations can cancel or reinforce one another. A set of parties may therefore cross the legislative majority threshold before its combined vote share reaches one half, leaving the complementary set with more votes but fewer seats. This configuration is a coalition inversion: a set of parties holds a seat majority while receiving less than half of all valid national votes cast for the legislature.

Kurrild-Klitgaard (2013) documents empirical cases of coalition inversions in Danish government formation. Miller (2014) establishes the more general result that inversions remain possible even when seats are allocated in a single national constituency, without an explicit seat threshold, and by a highly proportional formula. Because seats must be assigned in whole numbers, party seat shares will generally depart slightly from exact proportionality. Once positive and negative party-level deviations exist, some combinations of parties can reverse the vote and seat majorities. These results settle the existence question. They do not show how coalition inversions appear when national party seat totals are themselves assembled from many district elections.

Brazil provides such a setting. The Chamber of Deputies is not formed by applying one proportional allocation to national party votes. Its seats are allocated separately across the twenty-six states and the Federal District. District magnitudes and electoral weights vary, electoral rules and the units over which votes were pooled changed across the three elections, and party support is distributed unevenly across the country. Several institutional and geographical mediations therefore stand between a party's national vote share and its final national seat share (Nicolau 2006; Samuels and Snyder 2001; Bochsler 2026).

This institutional structure changes the sources of deviation without changing the national proportionality benchmark. I compare seat shares with shares of all valid federal-deputy votes, including votes for parties that obtain no seats. Restricting the denominator to votes for seat-winning parties would instead evaluate representation conditional on which parties entered the Chamber. The question here concerns the national outcome of the entire vote-to-seat process, including electoral exclusion. The benchmark therefore measures departures from national proportionality without presuming that those departures are unintended or unjustified.

Party-level deviations emerge from twenty-seven district-level translations operating under heterogeneous conditions, rather than only from rounding a single national allocation. The organizing question is therefore: when do these deviations combine so that a politically meaningful set of parties holds a legislative majority without a national vote majority? The contribution lies in showing how the resulting party-level seat differentials aggregate across coalitions and change majority status.

The Chamber contains an enormous number of mathematically possible party subsets, most of which have little political meaning. Realized cabinet coalitions provide the first empirical domain. Brazilian presidents normally distribute cabinet offices among several parties, and changes in ministerial participation create successive party sets during a legislative term (Abranches 1988; Figueiredo and Limongi 2000; Pereira and Mueller 2004). Each change may produce a new realized party set. Comparing that set's Chamber seats with the votes received by its members in the corresponding election shows when a cabinet coalition held a legislative majority without a national vote majority.

Cabinets are politically realized, but they are also selective and may combine parties separated by substantial ideological distance. A second domain asks whether the same aggregation problem appears among potential coalitions disciplined by ideological proximity. The baseline uses minimal connected winning coalitions: contiguous blocs in the election-year left-right order that hold a seat majority and cannot be narrowed further while retaining it (Axelrod 1970). Exact connectedness is a deliberately restrictive baseline. Since party positions are estimated and actual coalitions need not include every intermediate party, a one-gap sensitivity extension permits one intermediate party to be omitted. The one-gap domain therefore provides a nested relaxation of the no-gap requirement while preserving ideological proximity as the organizing restriction. Across both domains, the relevant quantity is the coalition's accumulated seat differential, defined as the difference between its seats and the number implied by its national vote share. That differential admits an exact algebraic decomposition into a within-district term and a between-district term. The terms make cases comparable by showing whether the two components reinforce or offset one another.

The empirical pattern spans both coalition domains and all three election cycles. Realized cabinet inversions and inversions among minimal connected winning coalitions appear following the 2014, 2018, and 2022 elections. The 2022 cases include a compact right-side bloc with 45.26 percent of the vote and 258 seats. Within-district and between-district components reinforce one another in some inversions and offset one another in others.

The remainder of the article follows the order of these transformations. Section 2 explains how Brazil's multidistrict open-list system converts district votes into the national party seat totals that coalitions later combine. Section 3 defines coalition inversion and the comparative decomposition, and Section 3.2 describes the data. Sections 4–4.3 move from party-level deviations to realized cabinet coalitions, ideologically constrained coalitions, and the comparison of decomposition patterns. Section 5 discusses the implications, and the final section concludes.

2 Brazil's multidistrict open-list proportional system

Before coalition majorities can be compared, one institutional question must be answered: how do twenty-seven separate electoral contests produce the national party seat distribution that coalitions later combine? Elections to Brazil's Chamber of Deputies take place inside the states, not in a single national constituency. Each of the twenty-six states and the Federal District forms a separate electoral circumscription for federal-deputy elections[1]. Seats are allocated independently in these twenty-seven districts and together compose the 513-member Chamber. No national compensatory calculation revises the party totals once the district counts have been completed (Brasil 1993; Câmara dos Deputados 2026; Tribunal Superior Eleitoral 2021b).

The number of seats at stake varies sharply across districts. The constitutional floor is eight the ceiling is seventy. Eleven districts elect eight deputies, the median magnitude is ten, the mean is nineteen, and São Paulo alone elects seventy. This distribution matters because the mechanical environment faced by parties is usually much smaller than the national size of the Chamber suggests. This distribution matters because the mechanical environment faced by parties in most districts is much smaller than the national size of the Chamber suggests. Much of Brazilian proportional representation operates near the lower end of the district-magnitude range, where each seat is a relatively large indivisible share of the delegation and the effective barrier to representation tends to be higher. Larger districts offer a more permissive environment, but because seats are allocated independently across states, district-level representational shortfalls are not compensated at the national level.

Within each district, voters encounter an open-list election. They may enter the identifying number of an individual candidate or cast a vote for a party label. A candidate vote performs two functions at once. It contributes to the aggregate vote of the party, electoral coalition, or federation through which the candidate competes, and it raises that candidate's position within the relevant open list. A party-label vote contributes to the first calculation but not to the ranking among candidates. That is, votes first determine how many seats an electoral list receives. Nominal votes then determine which candidates within that list take the seats, subject to the individual eligibility requirements in force for the election. A candidate can therefore benefit from votes cast for copartisans or electoral partners, but cannot ordinarily occupy a seat unless the list to which the candidate belongs wins one (Nicolau 2006; Brasil 1997).

The count begins by aggregating all valid candidate and party-label votes credited to the relevant electoral unit in district d. Depending on the election, that unit may be a party, a proportional electoral coalition, or a party federation. The institutional differences among these arrangements

1. The Federal District is therefore one ordinary eight-seat district within this architecture.

are discussed below. The electoral quotient, conventionally denoted by QE, is obtained by dividing the district's valid proportional-election vote by its magnitude and applying the statutory rounding rule. The party quotient then divides the vote total of each electoral unit by QE and discards the fractional part:

$$QE_d = \text{round}_{\text{stat}}\left(\frac{V_d}{M_d}\right), \qquad QP_{\ell d} = \left\lfloor \frac{v_{\ell d}}{QE_d} \right\rfloor.$$

Here V_d is the total valid vote in district d, M_d is its district magnitude, and $v_{\ell d}$ is the vote credited to electoral unit ℓ. The party quotient determines the unit's initial seat allocation, subject to the candidate-level eligibility requirements applicable in that election (Brasil 1965, arts. 106–108).

Because the party quotient discards fractional remainders, this first stage normally leaves seats unallocated. The remaining seats are distributed sequentially through the statutory highest-averages procedure. If electoral unit ℓ has already been allocated $s_{\ell d}$ seats, its value for the next seat is

$$m_{\ell d} = \frac{v_{\ell d}}{s_{\ell d} + 1}.$$

The denominator includes the seat currently being allocated, so $m_{\ell d}$ is the unit's average number of votes per seat if it were to receive that additional seat. The seat goes to the eligible unit with the highest value, its seat total is updated, and the comparison is repeated until all seats in the district have been allocated. The resulting priority quotients follow the Jefferson–D'Hondt sequence $v_{\ell d}/1, v_{\ell d}/2, v_{\ell d}/3, \ldots$. The D'Hondt highest-averages sequence governs the allocation of remaining seats, but it operates after the electoral- and party-quotient stage and subject to the election-specific eligibility rules described below (Brasil 1965; Nicolau and Schmitt 1995, art. 109).

The elections differed in the legal thresholds governing access to seats and the remainder allocation. In 2014, there was no fixed candidate-level threshold, but only parties or proportional coalitions that reached the electoral quotient could compete for remaining seats (Tribunal Superior Eleitoral 2013, art. 186, §2). In 2018, candidates ordinarily needed at least 10 percent of the electoral quotient to occupy a seat, while all parties and coalitions could enter the remainder competition regardless of whether they had reached the quotient, provided an eligible candidate remained; once no such candidate remained, the final seats were allocated by highest averages without that nominal-vote minimum (Tribunal Superior Eleitoral 2017, arts. 7–10). Under the rules applied in the original 2022 allocation, the 10 percent candidate threshold remained for initial seats, while the first remainder stage required parties or federations to reach 80 percent of the electoral quotient and candidates to reach 20 percent; the final highest-averages stage retained the party-level requirement but not the 20 percent candidate threshold (Tribunal Superior Eleitoral 2021b, arts. 8–11). These changes affected eligibility for marginal seats rather than the basic allocation sequence described above.

Moreover, proportional electoral coalitions, or *coligações proporcionais*, were part of the 2014 and 2018 elections. They were formed within the electoral circumscription, which for the Chamber meant a state or the Federal District. A party could consequently enter different proportional arrangements in different states, subject to the legal restrictions in force for the election. These were not national alliances whose composition automatically carried across all twenty-seven contests (Nicolau 2006; Tribunal Superior Eleitoral 2013, 2017). Proportional coalitions member parties legally operated as a joint open list when votes were converted into seats. Candidate votes and party-label votes credited to any member entered the coalition's district total; the quotient and highest-averages calculations then assigned seats to the coalition as a whole. Those seats were occupied by the highest nominal vote-getters across the combined list, regardless of the member party to which they belonged. A candidate from one party could therefore be elected partly because votes cast for candidates or labels of another coalition member raised the joint total (Nicolau 2006; Tribunal Superior Eleitoral 2013, 2017). However, pooling did not make the parties disappear. Candidates remained affiliated with member parties, used party identifiers on the ballot, and campaigned in a setting where party labels and candidate numbers remained central. The coalition changed the pool within which votes were translated into seats; it did not convert its members into a single political organization. Nor did it promise a stable proportional return to each partner. The number of seats won by the joint list was collective, while the distribution of those seats among member parties depended on how their candidates ranked in nominal votes across the coalition.

This arrangement has a clear family resemblance to list apparentement because several parties combine votes before the seat total is settled. The canonical apparentement studied by Pukelsheim and Leutgäb (2013), however, is a two-level apportionment: the alliance first receives seats collectively, after which those seats are apportioned among its component lists. Brazil instead used a single joint open list, with candidate nominal votes selecting the winners across party boundaries. Vote pooling can improve the electoral position of allied lists, while the allocation of the resulting advantage among particular partners is highly sensitive to the configuration of votes and alliances and can acquire a lottery-like quality. In Brazil, too, vote pooling could improve the electoral prospects of an alliance or of particular member parties, but any member-party gains depended on the distribution of candidate votes within the joint list rather than on a fixed rule allocating bonus seats to coalition partners.

Constitutional reform ended proportional electoral coalitions before the 2022 general election, and party federations became the available form of multiparty pooling in proportional contests (Brasil 2017, 1995). Three federations competed that year: Federação Brasil da Esperança, joining PT, PCdoB, and PV; Federação PSDB Cidadania; and Federação PSOL REDE (Tribunal Superior Eleitoral 2026). Like the former coligações, a federation pooled its members' votes to determine how many seats its common list received, while nominal votes determined which candidates occupied those seats. The resulting seat totals of the constituent parties were therefore not independent proportional allocations from their own vote totals, but ex post outcomes of competition within the pooled list.

The principal difference from proportional electoral coalitions lies in the federation's durability and organizational scope. A federation operates nationally and, in principle, persists for at least four years, whereas the former proportional coalitions were temporary electoral arrangements. This persistence carries some procedural consequences inside Congress, including common representation for certain leadership and committee-allocation purposes, but it does not merge the constituent parties: they retain separate identities and organizational autonomy (Brasil 1995; Tribunal Superior Eleitoral 2021a, 2022). Federations therefore add another institutional mediation between party votes and the party-level seat totals ultimately observed in the Chamber, rather than replacing those parties as political actors. (Brasil 1995; Tribunal Superior Eleitoral 2021a, 2022, 2019).

National proportionality provides a common baseline against which to assess how Brazil's layered vote-to-seat process generates representational deviations and when those deviations become majority-changing as parties are recombined into coalitions. If v_i is the national valid vote total credited to party i and V is the corresponding total for all parties, including those that win no seats, the party's reference quota is $513v_i/V$ seats. Brazil's actual system never performs this national allocation.

Several mediations therefore stand between a party's national vote share and its national seat share. Within each district, finite magnitude, the quotient and highest-averages calculations, legal eligibility thresholds, and the number and relative strength of competing electoral units shape the conversion of votes into seats. Electoral pooling adds a further mediation by changing both the vote total on which seats are allocated and, through the common candidate list, the eventual distribution of those seats among member parties. These arrangements varied across districts in 2014 and 2018 and became nationally fixed through federations in 2022.

The national seat distribution then aggregates twenty-seven independently determined district outcomes without a compensatory tier. Chamber seats are apportioned by population, but the constitutional floor of eight and ceiling of seventy prevent exact population-to-seat equality across states (Samuels and Snyder 2001; Brasil 1993). Turnout and valid-vote rates also differ across elections and districts, so a seat can correspond to different numbers of valid votes in different states. Because parties distribute their support unevenly across the country, national vote shares can therefore be assembled from geographically different configurations with different representational consequences. Electoral geography interacts with district seat weights and local competitive conditions before the resulting party seats are summed nationally. The resulting disproportionality need not have a single source: allocation rules, district magnitude, malapportionment and electoral weight, electoral geography, and party competition constitute distinct channels through which party vote and seat shares can diverge (Bochsler 2026).

This layered translation occurs in an unusually fragmented party environment. Measured by the Laakso–Taagepera index, the effective numbers of electoral parties were 14.11 in 2014, 18.02 in 2018, and 12.34 in 2022; the corresponding effective numbers of parliamentary parties were 13.33, 16.41, and 9.91 (Laakso and Taagepera 1979). The exact number of allocation units was not identical to the number of party labels, especially when district coalitions or national federations pooled votes. Even so, each election produced a Chamber in which legislative majorities had to be assembled across numerous partisan units. Fragmentation makes the political consequences of modest party-level gains and losses depend on how those parties are combined.

The national party seat totals produced by this process enter a presidential system in which governing capacity ordinarily depends on multiparty arrangements. Brazilian coalitional presidentialism links the presidency to combinations of parties across cabinet and Congress: ministries are distributed among several parties, while executive–legislative coordination is organized through agreements among them (Abranches 1988; Figueiredo and Limongi 2000; Pereira and Mueller 2004). The Chamber assembled from twenty-seven district elections therefore supplies the parliamentary resources from which governing coalitions are constructed. Party-level representation is politically consequential not only in isolation, but also through the combinations of parties that can command a legislative majority.

National proportionality provides an analytical baseline for examining this second stage. The realized party seat totals incorporate the effects of rounding, highest averages, district magnitude, eligibility requirements, electoral pooling, apportionment, turnout, valid-vote variation, electoral geography, and district-level competition. Comparing those totals with national proportional quotas identifies the positive and negative representational deviations carried by each party without attributing them to any single mechanism. Once parties are combined into politically relevant coalitions, these deviations may reinforce or offset one another. The next section formalizes this aggregation and states the condition under which the accumulated deviation carries a vote-minority coalition across the Chamber's 257-seat majority threshold.

3 Coalition inversions, domains, and measurement

The national seat total received by each party need not coincide with the quantity implied by exact national proportionality. A party may finish with a small positive seat deviation, while another finishes with a small negative one. In isolation, either difference may have little political significance. Coalition formation changes the scale of the comparison because each member brings its deviation into the combined party set. Positive and negative deviations may cancel, but they may also reinforce one another. A set of parties can therefore cross the Chamber's 257-seat majority threshold before its combined vote reaches one half of the national total. The complementary set then has more votes but fewer seats. This is the basic mechanism of coalition inversion.

The possibility of coalition inversions under proportional representation is already established by Kurrild-Klitgaard (Kurrild-Klitgaard 2013) and Miller (Miller 2014). The present analysis extends this agenda to a distinct institutional setting, where national party seat totals emerge from twenty-seven independent district allocations and several additional electoral mediations, and to coalition domains defined by realized cabinet participation and ideological proximity.

A fragmented legislature permits an enormous number of mathematically possible party subsets. An unrestricted search would be exhaustive, but it would place politically remote combinations on the same footing as combinations that were realized in government or structured by recognizable political proximity. The coalition space is therefore restricted in two different ways. Observed cabinets identify realized governing party combinations. Ideological connectedness supplies a disciplined domain of potential coalitions. Taken together, they use coalition inversion as a lens on two different features of the political system: how governing coalitions recombine party-level representational advantages in practice, and where potential vote-minority seat majorities are located within the ideological structure of the legislature.

Cabinet coalitions are substantively meaningful because Brazilian presidents ordinarily govern through multiparty arrangements. Cabinet participation records a party combination that was not merely arithmetically possible but institutionally realized in executive government (Abranches 1988; Figueiredo and Limongi 2000; Pereira and Mueller 2004). The relevant representational question is, thus, whether the members of the party set holding ministries collectively possessed a Chamber majority despite receiving less than half of the valid federal-deputy vote in the election that produced that legislature.

This domain must be followed across the legislative term. Coalitional presidentialism is dynamic: parties enter and leave the government, changing the set represented in cabinet while the underlying election-year vote and seat distribution remains fixed. These transitions generate successive realized combinations of the same party-level differentials. Cabinet participation is used only to delimit those observable governing party sets. This provides a useful setting for studying coalition inversion because changes in inversion status can occur without any change in the underlying electoral result: what changes is the particular combination of parties across which the fixed vote-seat differentials are aggregated. The cabinet sequence therefore reveals whether, and for how long, politically realized recombinations of the same legislature cross the majority threshold without a corresponding vote majority.

Observed cabinets reveal which combinations entered government, but they cover only a small portion of the majority configurations permitted by the legislature. A systematic analysis of potential coalitional majorities therefore requires a larger domain. Ideological proximity restricts this combinatorial space and shows whether vote-minority seat majorities can be assembled from ideologically coherent blocs and where such seat-efficient majority regions lie in the party system. The ideological domain, thus, extends the inversion analysis to politically structured potential coalitions. I use Axelrod's connected-coalition criterion as the baseline: an admissible coalition must occupy a contiguous interval in the ideological ordering (Axelrod 1970). The criterion is not intended as a model of Brazilian government formation. It defines a politically structured domain of potential coalitions in which ideological distance constrains which party combinations are compared.

Within this domain, I focus on minimal connected winning coalitions. A connected coalition is minimal winning if it holds at least 257 seats but no proper connected subset also holds a majority. Minimality removes larger intervals that are winning only because they contain a smaller winning coalition and isolates the inclusion frontier at which connected party sets acquire majority status. The inversion question can then be asked directly at that frontier: which of these ideologically coherent seat majorities still fall short of a national vote majority? Exact contiguity is a demanding admissibility condition, particularly when party positions are estimated and neighboring parties may occupy similar locations. I therefore consider a nested one-gap relaxation that permits one party inside the coalition's ideological span to be omitted to assess the sensitivity of the ideological-coalition results to the exact no-gap requirement while preserving ideological proximity as the organizing principle.

3.1 Measurement

Let P denote the set of all parties receiving valid votes in a federal-deputy election, including those that win no seats. For each party $i \in P$, let v_i be its national total of valid candidate and party-label votes and s_i its Chamber seats. With $V = \sum_{i \in P} v_i$ denoting the total valid federal-deputy vote and $S = 513$, exact national proportionality assigns party i the quota

$$q_i = S\frac{v_i}{V}.$$

The party's seat differential is $d_i = s_i - q_i$. A positive differential means that the party receives more seats than its national vote share implies, while a negative differential means that it receives fewer. Since party seats and party quotas both sum to S, the differentials sum to zero. For a coalition $C \subseteq P$, its vote total, seat total, proportional quota, and seat differential are

$$v_C = \sum_{i \in C} v_i, \qquad s_C = \sum_{i \in C} s_i,$$

$$q_C = S\frac{v_C}{V}, \qquad d_C = s_C - q_C.$$

Because coalition votes, seats, and quotas are obtained by summing over members,

$$d_C = s_C - q_C = \sum_{i \in C} (s_i - q_i) = \sum_{i \in C} d_i.$$

The coalition's departure from exact national proportionality is produced by adding the positive and negative differentials of its member parties. A party with a negative contribution can belong to an inverted coalition, provided that the positive contributions elsewhere in the set more than offset it.

A coalition inversion occurs when the party set receives less than half of the national valid federal-deputy vote but controls at least 257 Chamber seats: $\frac{v_C}{V} < \frac{1}{2}$ and $s_C \geq 257$.

For a coalition receiving less than half of the national vote, the seat-majority condition can be written as $d_C \geq 257 - q_C$. The right-hand side is the number of seats separating the coalition's exact proportional quota from a Chamber majority. An inversion occurs when the coalition's accumulated seat differential is large enough to bridge that gap. Since C and its complement $\bar{C} = P \setminus C$ partition both the full election-year party set and the 513 seats, $\frac{v_C}{V} < \frac{1}{2},\ s_C \geq 257 \iff \frac{v_{\bar{C}}}{V} > \frac{1}{2},\ s_{\bar{C}} \leq 256$. The criterion therefore identifies a reversal between the coalition and its complement[2]. Throughout the article, V includes all valid federal-deputy votes rather than only votes cast for seat-winning parties or another restricted vote total.

Representation ratios express the same departures in relative rather than absolute terms (Taagepera and Laakso 1980). At the party and coalition levels,

$$R_i = \frac{s_i/S}{v_i/V} = \frac{s_i}{q_i}, \qquad R_C = \frac{s_C/S}{v_C/V} = \frac{s_C}{q_C} = 1 + \frac{d_C}{q_C} = \sum_{i \in C} \frac{q_i}{q_C} R_i.$$

A value of one denotes exact proportionality, a value above one denotes overrepresentation, and a value below one denotes underrepresentation. Parties receiving votes but no seats have $R_i = 0$.The ratios permit comparisons across parties and coalitions of different sizes. The differentials remain the more direct quantities for the majority-threshold question because they measure the relevant departure in seats.

The national coalition differential can be decomposed by adding and subtracting the seats that exact proportionality would assign to the coalition separately within each district:

$$d_C = \left[\sum_d \left(s_{Cd} - S_d \frac{v_{Cd}}{V_d}\right)\right] + \left[\left(\sum_d S_d \frac{v_{Cd}}{V_d}\right) - \left(S \frac{v_C}{V}\right)\right].$$

The first term compares the coalition's actual seats in each district with the number implied by its vote share in that district. I denote it the within-district component, A_C. The other component is the between-district component, B_C. It's first term sums the coalition's proportional quota in each district. The second is the quota implied by the coalition's share of the national valid vote. These quantities need not coincide because districts differ in the number of seats they contribute relative to the number of valid votes cast in them. Thus, B_C depends on the geographic distribution of coalition support across districts with different electoral weights. Support concentrated in districts whose share of Chamber seats exceeds their share of the national valid vote raises the sum of district quotas relative to the national quota, while support concentrated in districts with the opposite weighting lowers it.

Thus, $d_C = A_C + B_C$. For comparisons across coalitions of different electoral sizes, the two components can also be expressed relative to the coalition's proportional quota:

$$R_C - 1 = \frac{A_C}{q_C} + \frac{B_C}{q_C}.$$

This expresses the within- and between-district contributions to the coalition's relative departure from proportionality.

In Brazil, A_C can reflect the mechanisms that operate within each district when votes are converted into seats: finite district magnitude, the electoral-quotient and highest-averages calculations, the year-specific eligibility rules for parties and candidates, and the pooling of votes through proportional electoral coalitions in 2014 and 2018 and party federations in 2022. These mechanisms determine how the coalition's actual district seats depart from its district-level proportional quotas.

2. This structure differs from Siaroff's (Siaroff 2003) distinction between manufactured and spurious majorities. A manufactured majority occurs when the vote-leading party has less than half of the vote but receives a seat majority; a spurious majority occurs when the seat-majority party received fewer votes than another party. Coalition inversion necessarily has the stronger reversal structure. Because C and its complement $\bar{C}$ exhaust both votes and seats, $v_C/V < 1/2 < s_C/S$ implies that $\bar{C}$ has a vote majority but a seat minority. The outcome is therefore analogous simultaneously to a spurious majority for C and an artificial minority for $\bar{C}$. Siaroff's units, however, are parties or electoral alliances treated as parties, whereas the objects considered here are sets of parties. I therefore retain the term *coalition inversion*.

By contrast, B_C arises from differences in electoral weight across districts and the geographic distribution of coalition support: district apportionment, turnout, and valid-vote variation determine how many seats correspond to a given number of votes in each district, while coalition geography determines whether its support is concentrated in relatively more or less heavily weighted districts. Neither component isolates the independent effect of any one mechanism; the decomposition locates the coalition differential within or between districts rather than providing a causal decomposition of its institutional sources (Bochsler 2026).

The primary ideological universe is the set of parliamentary parties,

$$P^+ = \{i \in P : s_i > 0\}.$$

National proportionality is evaluated over P, while potential parliamentary coalitions are constructed over P^+. Thus, votes for parties without seats remain part of V and of party-level proportionality accounting, but those parties cannot constitute or interrupt a primary ideological coalition. This restriction does not alter observed cabinet membership or its election-year translation.

Axelrod's connected-coalition criterion provides the baseline restriction (Axelrod 1970). Let $p_1, \ldots, p_n$ denote the existing election-year ideological order restricted to P^+, preserving the relative ordering of represented parties without re-estimating their positions. A coalition $C \subseteq P^+$ is connected when its members form a contiguous interval in this filtered order. A connected coalition is minimal winning when $s_C \geq 257$ and no proper connected subset retains a Chamber majority. For an interval, removing either endpoint must therefore leave fewer than 257 seats.

The one-gap relaxation also uses the filtered parliamentary order. For any nonempty $C \subseteq P^+$, let $\text{span}(C)$ contain all represented parties between its leftmost and rightmost members, and define

$$g(C) = |\text{span}(C) \setminus C|, \qquad \mathcal{D}_k = \{C \subseteq P^+ : C \neq \varnothing,\ g(C) \leq k\}.$$

The exact-connected domain is $\mathcal{D}_0$. The domain $\mathcal{D}_1$ additionally permits one represented party inside the span to be omitted. A party with no seats contributes no gap in either primary domain. In each $\mathcal{D}_k$, a winning coalition is minimal if no proper subset that is also in $\mathcal{D}_k$ has at least 257 seats. Minimality is recomputed separately for $k = 0$ and $k = 1$; changing admissibility can change which coalitions are minimal. Inversions are then identified among these minimal winning coalitions using $v_C/V < 1/2$, with the same all-valid-vote denominator.

As a sensitivity analysis, I repeat both searches in the full election-year order over P, including parties receiving votes but no seats. In that universe, such parties can be coalition members and can interrupt connectedness. Admissibility and minimality are recomputed within that order. Only the ideological universe changes: votes, seats, quotas, differentials, representation ratios, and the decomposition retain the same definitions and national vote denominator. Appendix D compares the results.

3.2 Data and operationalization

Vote and seat data for the 2014, 2018, and 2022 federal-deputy elections come from the Tribunal Superior Eleitoral, accessed through the `electionsBR` package (Meireles, Silva, and Costa 2016).[3] For each party, the national vote total combines valid candidate votes and party-label votes across the twenty-seven districts, while the seat total counts deputies elected under its election-year label.[4]

3. All 2022 results refer to the official seat vector in the frozen dataset, prior to subsequent judicial retotalizations.

4. For 2014, the vote measure is `QT_VOTOS_NOMINAIS + QT_VOTOS_LEGENDA`. For 2018 and 2022, it is `QT_VOTOS_NOMINAIS_VALIDOS + QT_TOTAL_VOTOS_LEG_VALIDOS`.

This attribution also applies to parties that competed through proportional electoral coalitions in 2014 and 2018 or federations in 2022. The resulting party totals connect the district electoral outcomes to the party identities observed in cabinet participation and ideological classifications.

Cabinet composition is reconstructed from the Brazilian Wikipedia cabinet-member lists for the Dilma Rousseff, Michel Temer, Jair Bolsonaro, and Lula III administrations. These records identify each office, incumbent, party affiliation, and dates of entry and exit. A cabinet period lasts as long as the set of parties holding ministries or ministry-level offices remains unchanged. The entry or exit of a party begins a new period.

The 2014 election is linked to the cabinet sequence from January 1, 2015 through December 31, 2018, and the 2018 election to the sequence from January 1, 2019 through December 31, 2022. The 2022 election is linked to the observed Lula III period, from January 1, 2023 through March 19, 2026. Periods are first delimited using the party labels observed in cabinet. These labels are then translated into those used in the corresponding election, and coalition votes and seats are obtained by summing the member-party totals. Renames are matched to earlier labels, while parties formed through fusion are mapped to their constituent election-year parties.[5] Appendix B reports the cabinet sequence used in the analysis, and the replication materials document the party translations.[6]

The ideological orders for 2014 and 2018 use the party classifications of Bolognesi, Ribeiro, and Codato (2023). For 2022, I use the later 2018–2022 classification, drawing on the 2025 data release and associated article (Bolognesi et al. 2025; Bolognesi et al. 2026). Each classification supplies the election-year party order used to construct the coalition domains defined above.

4 Results

Before turning to coalitions, I first examine the party-level seat differentials d_i and representation ratios R_i produced in each election, which describe the absolute and relative departures from national proportionality that are subsequently combined at the coalition level.

In the 2014 election PMDB received 8.13 seats above its national proportional quota, the largest positive differential that year. PSD and PTB received another 4.55 and 4.37 seats above their quotas, respectively. These gains coexisted with deficits of 4.43 seats for PSDB and 4.20 for PSOL.

In 2018, PP, PR, and MDB received positive differentials of 8.76, 5.72, and 5.60 seats, while PSL and NOVO fell 7.70 and 6.35 seats below their national quotas. PSL is especially instructive: it received the largest national vote total but also the largest negative seat differential. This is not paradoxical in a multidistrict system. A large national vote total can be assembled disproportionately in districts where votes carry less weight in the national seat distribution, and party size magnifies the absolute consequence of even moderate underrepresentation. Electoral size therefore helps determine the magnitude of a party's contribution, while the direction of that contribution depends on how its votes are translated across districts.

In the 2022 election positive differentials are even more concentrated among largest parties: PL received 13.78 seats above its national proportional quota and UNIÃO another 11.10, while PT and PP gained 6.94 and 6.24 seats, respectively. These large absolute surpluses reflect both electoral size and overrepresentation. Since $d_i = q_i(R_i - 1)$, a party with a large proportional

5. Periods 2021.3 and 2022.1 are separated by the February 8, 2022 fusion of DEM and PSL into UNIÃO. When linked to the 2018 election, UNIÃO maps back to DEM and PSL, leaving the party set used for the vote-seat comparison unchanged. They are therefore combined in the analysis.

6. The replication materials are available at https://anonymous.4open.science/r/electoral-inversion-B6B5/README.md.

quota can contribute many seats to the positive or negative side of the national differential even when its relative departure from proportionality is moderate. In 2022, the largest parties happened predominantly to lie on the positive side of this relationship, placing substantial seat surpluses inside many of the coalition combinations available in the new Chamber.[7]

Figure 1 shows that electoral size structures the scale of representational departures without fixing their direction. Relative representation is considerably more dispersed among small parties: at low vote shares, parties range from receiving no seats to substantial overrepresentation, while larger parties generally lie closer to $R_i = 1$. Yet the side of proportionality on which larger parties fall changes across elections. In 2014, PT and PSDB were underrepresented while PMDB, of comparable size, was overrepresented; in 2018, the largest party, PSL, was also underrepresented. The 2022 profile differs sharply: every party receiving more than 5 percent of the national vote was overrepresented, while those between 3 and 5 percent were underrepresented. The national profiles therefore show no consistent tendency for larger or smaller parties to be overrepresented. Party size instead conditions how a given relative departure translates into an absolute seat differential.

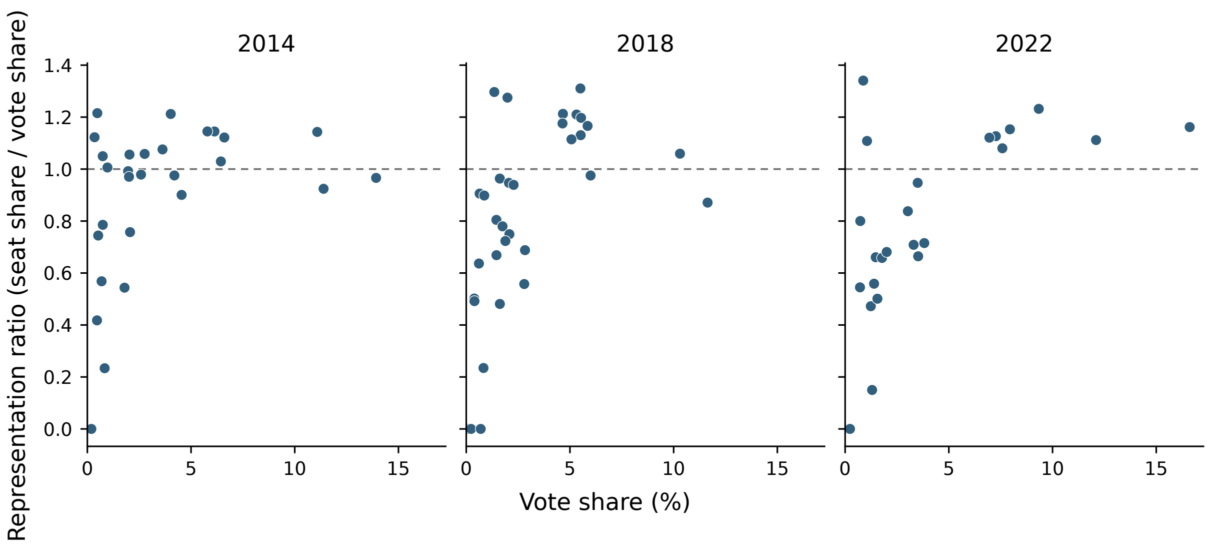

Figure 1: Party-level representation profiles, 2014, 2018, and 2022

Notes: Each point is a party. $R_i = 1$ denotes exact proportionality; parties receiving votes but no seats appear at zero.

The party profiles also clarify why representation ratios alone do not determine which parties contribute most to a coalition's seat surplus. Coalition representation is not determined by the number of overrepresented parties, nor by the largest representation ratio among its members. Each party enters in proportion to its electoral size: $R_C = \sum_{i \in C} \frac{q_i}{q_C} R_i$, and $d_C = \sum_{i \in C} d_i$. Thus, a strongly overrepresented small party may contribute relatively few seats, while a more moderately overrepresented large party may account for a substantial share of the coalition's surplus. In 2022, for example, PV had the higher representation ratio, $R_i = 1.341$, but contributed only 1.52 seats above quota; PL's lower ratio of 1.162 translated into a 13.78-seat surplus. Whether a coalition

7. The complete party-level vote shares, quotas, differentials, and representation ratios can be found in the replication repository.

reaches a majority therefore depends on the quota-weighted combination of these party-level advantages and losses, and on whether their resulting differential is large enough to bridge the remaining distance to 257 seats.

4.1 Cabinet coalitional inversions

Four of the twenty-three cabinet periods combine a Chamber majority with less than half of the national federal-deputy vote. They occur under all four administrations, but differ in their distance from a vote majority, the size of their seat advantage, and their duration. Table 1 reports these periods and the within-district and between-district components of their differentials.

Table 1: Observed cabinet coalition inversions and their district components

Election	Period	Days	Vote %	Seats	d_C	A_C	B_C	Parties
2014	2016.2	2	46.54	259	20.271	15.742	4.529	PCdoB, PDT, PMDB, PR, PSD, PT, PTB
2014	2017.1	100	49.91	266	9.952	11.268	-1.316	DEM, PMDB, PP, PPS, PSB, PSD, PSDB, PV
2018	2021.3/2022.1	238	47.25	257	14.624	19.069	-4.445	DEM, PATRIOTA, PP, PR, PRB, PSC, PSD, PSDB, PSL
2022	2023.1	255	48.89	263	12.204	13.196	-0.992	MDB, PCdoB, PDT, PSB, PSD, PSOL, PT, REDE, UNIÃO

Notes: Votes, seats, and party labels refer to the corresponding election. A_C and B_C are the within-district and between-district components.

The first inversion emerges during the contraction of Dilma Rousseff's cabinet in April 2016. After PRB and PP leave, the remaining parties retain 259 seats with 46.54 percent of the 2014 vote. Exact national proportionality would assign them 238.73 seats, leaving them 18.27 seats short of a Chamber majority. Their 20.27-seat differential, corresponding to $R_C = 1.085$, is therefore sufficient to preserve the majority. This configuration, period 2016.2, lasts two days and ends with PSD's departure.

The later Temer inversion begins after PTB leaves the cabinet in December 2017. The remaining parties retain 266 seats with 49.91 percent of the vote. Their proportional quota is 256.05 seats, only 0.95 seats below the majority threshold. A differential of 9.95 seats and $R_C = 1.039$ therefore produce a larger seat majority than in the Dilma episode despite a substantially smaller representational advantage. The comparison shows that the political consequence of a coalition's differential depends on its proportional distance from 257 seats: a party set already close to a vote majority requires much less overrepresentation to cross the legislative threshold. This configuration persists for 100 days.

The cabinet sequence linked to the 2018 election reveals a different relationship. Following NOVO's departure, PP's entry in August 2021 brings the cabinet party set to exactly 257 seats with 47.25 percent of the vote. Its proportional quota of 242.38 seats leaves a 14.62-seat gap, exactly matched by its differential, with $R_C = 1.060$. This configuration continues for 238 days, from August 4, 2021 through March 29, 2022. PSC's departure then reduces the coalition from 257 to 250 seats and ends the inversion. Yet PSC itself was underrepresented. Removing it therefore lowers the coalition's proportional quota by more than it lowers its actual seat total, increasing the remaining coalition's differential from 14.62 to 16.60 seats. The coalition is thus more overrepresented relative to its vote share, but no longer holds a Chamber majority. This distinguishes overrepresentation from inversion: the former compares a coalition's seats with its proportional quota, whereas the latter requires a coalition with less than half of the vote to cross the 257-seat majority threshold.

The initial Lula III cabinet provides the converse transition. Its members hold 263 seats with 48.89 percent of the 2022 vote. Their proportional quota is 250.80 seats, so a 12.20-seat differential, corresponding to $R_C = 1.049$, more than covers the 6.20 seats separating the coalition from a majority. After 255 days, the entry of PP and Republicanos expands the coalition to 350 seats and 63.79 percent of the vote. Whereas PSC's departure ends the Bolsonaro-period inversion by removing the seat majority, this expansion ends the Lula-period inversion by supplying a vote majority as well. Across these cases, the underlying election-year vote–seat distribution remains fixed. Inversion status changes because cabinet entry and exit alter both the coalition's proportional distance from 257 seats and the party differentials accumulated within it.

The within-district component is positive in every cabinet inversion, but the pattern extends to the entire cabinet sequence: $A_C > 0$ in all twenty-three periods, whereas $B_C > 0$ in only six. Cabinet composition helps explain the difference. Parties that enter cabinet are larger on average than parties that never do, with mean vote shares of 5.63 versus 0.91 percent in 2014, 5.05 versus 1.85 percent in 2018, and 5.80 versus 1.72 percent in 2022. Larger parties are also usually on the positive side of the within-district component. Among parties receiving at least 5 percent of the national vote, $A_i > 0$ for six of seven in 2014, eight of nine in 2018, and all seven in 2022.[8] In every distinct cabinet composition, the positive A_i contributions of members above this 5 percent benchmark exceed the combined negative contributions of the remaining members. Cabinet coalitions therefore repeatedly accumulate a positive within-district component. No equivalent pattern appears for B_C, which may reinforce or offset that advantage depending on where coalition support is distributed across districts.

The four inverted cabinet periods differ in how this positive A_C interacts with between-district weighting. In 2016.2, $A_C = 15.74$ raises the coalition from its proportional quota of 238.73 to 254.47 seats, still short of a Chamber majority. The additional $B_C = 4.53$ is therefore needed to cross the 257-seat threshold. In 2017.1, the 2018-linked configuration, and 2023.1, A_C alone is already sufficient to cross the threshold, while negative B_C reduces the resulting seat advantage. The largest offset occurs in the 2018 inversion, where $A_C = 19.07$ and $B_C = -4.45$, leaving exactly 257 seats. The common feature of the cabinet inversions is thus the positive within-district component; between-district weighting determines whether that advantage is reinforced or reduced and, in 2016.2, whether it is enough to produce an inversion.

The full cabinet sequence shows that overrepresentation is much more common than inversion. Figure 2 shows that twenty of the twenty-three periods have $R_C > 1$; the only three underrepresented configurations occur during the Bolsonaro administration. Most overrepresented cabinets do not reverse majority status: some remain below both 50 percent of the vote and 257 seats, while others exceed both. The four inversions are the cases in which these two thresholds diverge, with the coalition holding at least 257 seats while remaining below 50 percent of the vote.

Measured over time, inversions occupy 102 of the 1461 covered days following the 2014 election, 238 of 1461 following the 2018 election, and 255 of 1174 in the observed period following the 2022 election. The mismatch therefore extends beyond the brief 2016 transition to configurations maintained for several months. Its appearance and disappearance follow changes in cabinet membership: party entry or exit can move a coalition across the seat-majority threshold, the vote-majority threshold, or both. The same electoral distribution can therefore sustain different majority relations as its parties are recombined in government.

8. The parties above 5 percent with positive A_i are PSDB, PMDB, PP, PSB, PSD, and PR in 2014; PT, PSDB, PSD, MDB, PSB, PP, PR, and PRB in 2018; and PL, PT, UNIÃO, PP, PSD, MDB, and Republicanos in 2022. The exceptions are PT in 2014 and PSL in 2018.

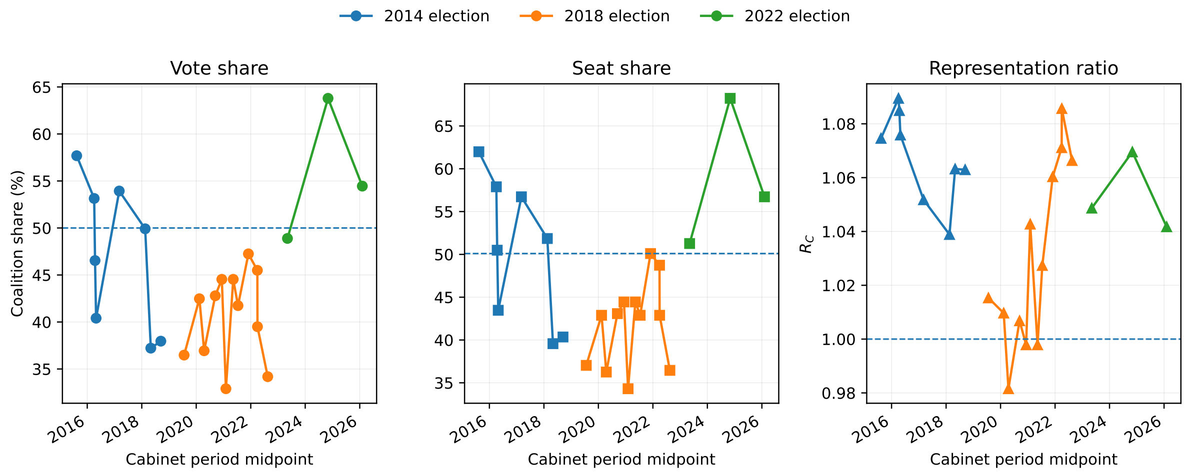

Figure 2: Observed cabinet-period coalition vote shares, seat shares, and representation ratios

4.2 Ideologically constrained potential coalitions

The exact-connected baseline yields inversions in all three elections: four of eight minimal connected winning coalitions invert in 2014, one of eight in 2018, and two of four in 2022. These coalitions are constructed over 28, 30, and 23 seat-winning parties, respectively. Table 2 reports minimal-majority and inversion counts and the strongest inversion under each admissibility condition. Figure 3 locates the exact-connected inversions within the ideological interval space.

Table 2: Minimal winning coalitions by ideological domain

Election	k	Minimal majorities	Inversions	Strongest minimal inversion
2014	0	8	4	PTB–PR (47.45%; 257 seats)
2014	1	87	46	PTB–PR (47.45%; 257 seats)
2018	0	8	1	PT–PSDB (49.95%; 267 seats)
2018	1	108	42	PCdoB–PODE, omitting PSDB (47.59%; 257 seats)
2022	0	4	2	PP–PL (45.26%; 258 seats)
2022	1	39	12	PP–PL (45.26%; 258 seats)

Notes: The ideological order contains seat-winning parties. k counts omitted parties inside a coalition's ideological span. Minimality is defined within each stated domain. Vote shares use all valid federal-deputy votes. The strongest minimal inversion has the lowest national vote share; exact ties use coalition size and then canonical membership order.

The four 2014 connected inversions contain twelve to fifteen parties. PSB–PTN begins on the center-left; the other three begin at parties classified as center-right, and all extend into the right. None includes PT, but every case combines PMDB and PSD with PSDB. The strongest inversion, measured by the lowest vote share, is the thirteen-party PTB–PR interval: 47.45 percent of all valid votes yields exactly 257 seats, a quota of 243.40 seats, and a differential of 13.60 seats. Its representation ratio is $R_C = 1.056$.

In 2018, PT–PSDB is the only exact-connected minimal inversion. Its fourteen parties receive 49.95 percent of all valid votes and hold 267 seats. The proportional quota is 256.25 seats, the differential is 10.75 seats, and $R_C = 1.042$. The other seven minimal connected majorities have vote majorities, including both overrepresented and underrepresented combinations. This election therefore supplies a narrow vote-minority inversion under exact connectedness, alongside the observed PSC-inclusive cabinet inversion.

In 2022, the non-inverted minimal majorities PT–PP and PSB–PSC begin on the left and center-left and contain fifteen and sixteen parties, respectively. They retain seat majorities despite negative differentials of 9.36 and 15.10 seats. The inversions occur in the ten-party MDB–UNIÃO interval, spanning the center-right through the far right, and the seven-party PP–PL interval, confined to the right and far right.

MDB–UNIÃO receives 49.22 percent of the vote, placing its proportional quota at 252.49 seats, 4.51 below the majority threshold. Its 12.51-seat differential and $R_C = 1.050$ yield 265 seats. PP–PL receives 45.26 percent of the vote and has a quota of 232.19 seats, leaving a 24.81-seat shortfall from 257. Its larger differential of 25.81 seats, corresponding to $R_C = 1.111$, yields a smaller majority of 258 seats. PL and UNIÃO contribute 13.78 and 11.10 seats above quota to this latter coalition, which also includes underrepresented members. The contrast shows why the size of a seat majority cannot be read directly from the extent of overrepresentation: a coalition with less electoral support requires a larger advantage to reach the same threshold. Table 3 reports the differentials and district components of all seven minimal connected inversions; full $k = 0$ minimal-winning compositions appear in Appendix C.

Allowing one party inside a span to be omitted changes both admissibility and minimality. All twenty exact-connected minimal majorities remain admissible and winning, but only six remain minimal in $\mathcal{D}_1$. The one-gap domain contains 46 minimal inversions among 87 minimal majorities in 2014, 42 among 108 in 2018, and 12 among 39 in 2022. These are overlapping potential configurations under different restrictions, not probabilities of government formation.

In 2014, the relaxation broadens the ideological distribution of inversions: their leftmost members range from PSOL through PMDB, and seven include PT. The strongest inversion remains the same thirteen-party PTB–PR coalition, with 47.45 percent of the vote and 257 seats. Since every coalition member has positive seats, this exact-threshold coalition has no proper winning subset and remains minimal when one gap is allowed.

The strongest 2018 one-gap case is PCdoB–PODE omitting PSDB. The full span contains sixteen parties and holds 287 seats with 53.59 percent of all valid votes. Omitting PSDB removes thirty seats, leaving fifteen parties with exactly 257 seats and 47.59 percent of the vote. Because PSDB is underrepresented, this exclusion raises the differential by 0.76 seats, from 12.10 to 12.86. The coalition retains its seat majority while losing its vote majority. Unlike PSC's departure from the Bolsonaro-period cabinet, this exclusion leaves the coalition at the seat-majority threshold. The one-gap search thus yields a larger vote deficit than the PT–PSDB exact-connected inversion.

In 2022, all twelve one-gap minimal inversions contain only parties classified as center-right, right, or far right, and none includes PT. They contain seven to eleven parties, with a median of nine. The strongest remains the same seven-party PP–PL interval as in $k = 0$, with 45.26 percent of the vote and 258 seats. Three other minimal inversions occupy the wider MDB–UNIÃO span with one party omitted. The right-side concentration therefore extends beyond the single coalition with the lowest vote share.

The strongest 2014 and 2022 inversions retain both their membership and their vote–seat outcomes under the one-gap relaxation, although the broader set of minimal coalitions changes in every election. The all-party sensitivity analysis preserves their endpoint spans but removes the 2018 exact-connected inversion: including PMB in PT–PSDB adds votes but no seats, giving the interval a vote majority. The PTB–PR and PP–PL member sets require a one-gap allowance in that universe because they omit PPL and DC, respectively. Appendix D reports the full comparison.

Table 3: Exact-connected ($k = 0$) minimal inversions and seat-differential decomposition

Election	Start	End	Parties	Vote %	Seats	d_C	A_C	B_C
2014	PSB	PTN	12	48.92	260	9.032	10.509	-1.477
2014	PTB	PR	13	47.45	257	13.601	12.507	1.094
2014	PT DO B	PSDC	15	48.84	257	6.442	7.153	-0.711
2014	SOLIDARIEDADE	PSL	15	48.84	257	6.462	7.473	-1.011
2018	PT	PSDB	14	49.95	267	10.751	6.568	4.183
2022	MDB	UNIÃO	10	49.22	265	12.507	2.953	9.554
2022	PP	PL	7	45.26	258	25.813	23.321	2.492

Figure 3 places these cases in the full exact-connected interval space. Blue cells reach a Chamber majority with a vote majority, orange cells are inversions, and red cells are minimal inversions. The 2014 cases occupy several center-to-right spans. The 2018 panel contains the PT–PSDB inversion, with a vote share just below one half. The 2022 inversion region is on the right side of the ideological order. The figure concerns the exact-connected baseline, $k = 0$.

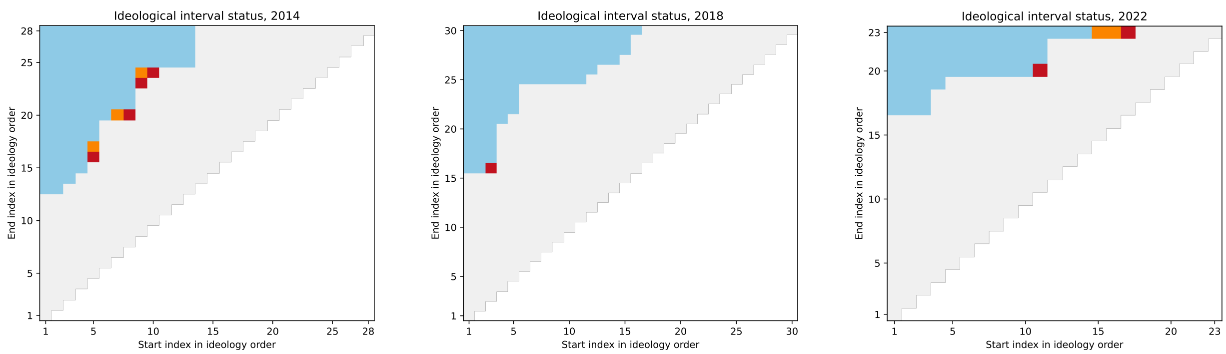

Figure 3: Exact-connected ($k = 0$) ideological interval status by election

Notes: Each matrix enumerates exact-connected $k = 0$ intervals. Start and end indices refer to the election-year ideological order of seat-winning parties. The one-gap $k = 1$ domain is not plotted.

In 2022, the inversion region lies at the right edge of the party order. MDB–UNIÃO spans a relatively broad center-right-to-far-right interval, whereas PP–PL is already a narrow right-side majority without a vote majority. Exact-connected minimal winning intervals that include PT remain broad vote-and-seat majorities rather than inversions. Appendix E documents the corresponding distinction for observed cabinets, which are sparse selections from wider ideological spans rather than the connected intervals plotted here. The next subsection compares the within-district and between-district components underlying these different combinations.

4.3 Comparing decomposition patterns across coalition domains

The member-party decomposition helps explain the persistent positive within-district component of cabinet coalitions. Cabinets repeatedly combine large parties carrying substantial positive A_i contributions. In the 2018 cabinet inversion, for example, PP and PR contribute 7.32 and 6.37 seats to A_C. In the initial 2023 cabinet, UNIÃO and PT contribute 9.36 and 9.07 seats. Because these contributions are summed when parties enter the same cabinet, the selection of such parties systematically raises the coalition's within-district component. This pattern extends beyond the inverted configurations: A_C is positive in all twenty-three observed cabinet periods.

The between-district component behaves differently because its sign depends on the geographic distribution of each party's support across districts with different electoral weights. Large parties can therefore contribute substantially in either direction. PSL contributes -5.28 seats to B_C in the 2018 cabinet inversion, while PT contributes -2.13 in the initial 2023 cabinet despite its large positive contribution to A_C. Between-district weighting is positive in only six of the twenty-three cabinet periods. The same parties that build the within-district advantage therefore need not provide a corresponding between-district advantage.

The positive cabinet A_C values are also accumulated across many districts rather than produced by only a few exceptional state results. Among the four distinct inverted cabinet configurations, between eighteen and twenty-two of the twenty-seven districts contribute positively to A_C. The three largest positive district contributions account for between 26.3 and 45.1 percent of the positive total, and between twelve and fifteen districts are required to account for 90 percent. The balance differs across configurations, but in each case the national within-district advantage combines positive contributions from much of the country.

The minimal connected inversions share the cabinets' positive within-district component, but its relative importance varies. In 2022, MDB–UNIÃO has $A_C = 2.95$ and $B_C = 9.55$: both components are positive, with between-district weighting providing the larger share of its 12.51-seat differential. PP–PL instead has $A_C = 23.32$ and $B_C = 2.49$. In 2018, the PT–PSDB inversion combines $A_C = 6.57$ with $B_C = 4.18$. The same inversion classification therefore covers substantially different balances between the components. Under the one-gap relaxation, three minimal inversions have a negative within-district component: two in 2014 and one in 2022. Their positive between-district terms more than offset that shortfall.

In the all-party sensitivity analysis, MDB–UNIÃO instead combines $A_C = -1.29$ with $B_C = 9.88$, while retaining its inversion status (Appendix D). Positive A_C in all minimal connected inversions in the baseline is therefore an empirical result for that domain, not a mathematical requirement for inversion.

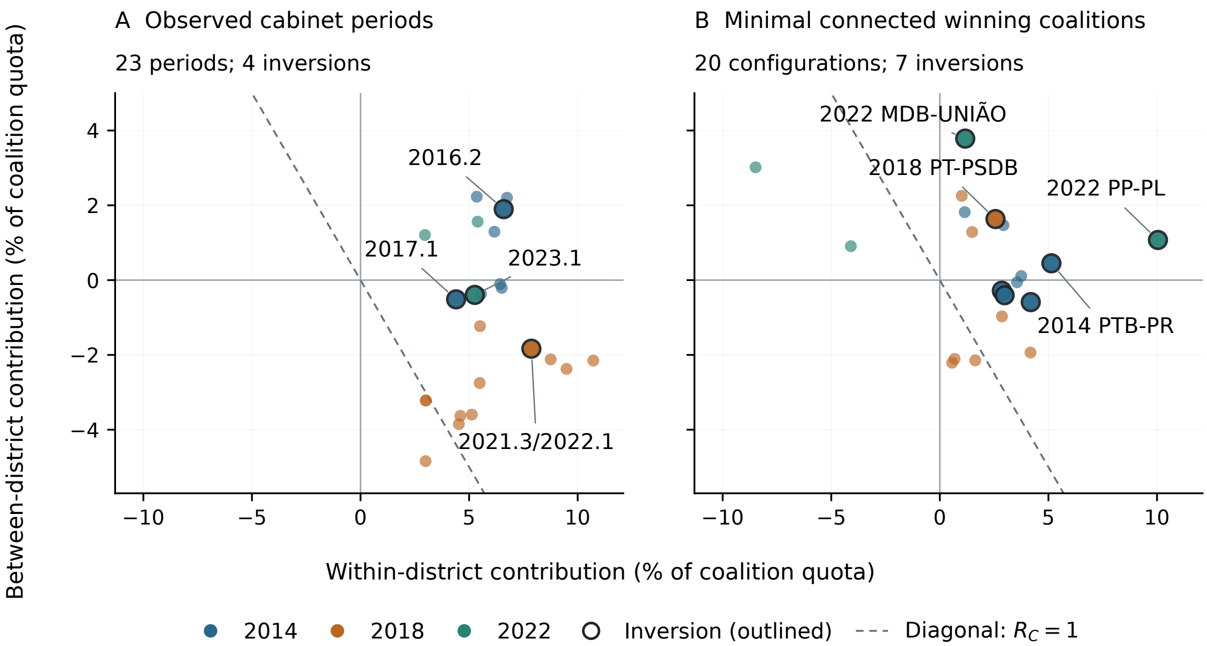

Figure 4: Within-district and between-district contributions across coalition domains

Notes: Components are expressed as percentages of each coalition's proportional quota. Colors identify election years and outlined points are coalition inversions. Panel A reports cabinet periods. Panel B reports minimal connected winning coalitions ($k = 0$). The dashed line marks $R_C = 1$.

Figure 4 compares the components after accounting for coalition size. Panel B contains twenty minimal connected winning coalitions, seven of which invert. Every observed cabinet configuration has a positive within-district component, whereas the ideological majorities occupy both sides of that boundary. The inverted connected coalitions are all on its positive side. Cabinet inversions are not concentrated at the largest normalized within-district values: several non-inverted cabinets have larger values. Across domains, majority status depends on both the components' sum and the coalition's proportional distance from 257 seats.

5 Discussion

The analysis compares coalition inversions across two politically defined domains in a proportional system institutionally different from the principal cases in the existing literature. The first domain follows the parties actually represented in cabinet as composition changes during a mandate. The second enumerates ideologically constrained potential coalitions, first under exact connectedness and then under a nested one-gap relaxation. The same party vote–seat distribution can therefore be evaluated under realized and structured potential recombinations.

The distinction matters because coalition inversion is configuration-dependent. In 2018, the 238-day PSC-inclusive cabinet period and the PT–PSDB interval both invert, but they combine different party sets. Relaxing exact connectedness admits a stronger inversion, PCdoB–PODE omitting PSDB, with 47.59 percent of all valid votes and 257 seats. The strongest 2014 and 2022 ideological inversions retain their PTB–PR and PP–PL compositions and outcomes under the same relaxation. Observed cabinets and ideologically constrained potential coalitions therefore reveal different majority configurations within the same electoral distribution.

The party representation profiles also show why coalition inversion should not be reduced to one stable party-size bias. In 2014, overrepresentation and underrepresentation are distributed across parties of different sizes. In the 2022 data, larger parties are more consistently overrepresented, but coalition outcomes still depend on which positive and negative party differentials are combined. The political consequence of a fixed electoral result is realized through the party sets that organize legislative and governing majorities.

At the coalition level, R_C provides the bridge between party representation and majority status. It is a quota-weighted aggregation of the member-party ratios, not a simple count of overrepresented parties. The cabinet timeline shows that relative advantage is common: 20 of 23 party sets have $R_C > 1$, but only four invert. Majority-changing advantage depends jointly on the size of $R_C - 1$ and on q_C, because the latter fixes how many surplus seats are needed to reach 257. The contrast between the MDB–UNIÃO interval, whose quota is closer to a majority, and the more strongly overrepresented PP–PL interval makes this distinction especially clear.

The decomposition of the exact-connected domain adds a second comparative result. The within-district component is positive in all four observed cabinet inversions and all seven minimal connected inversions. Between-district weighting reinforces it in the PTB–PR, PT–PSDB, MDB–UNIÃO, and PP–PL ideological cases and in the 2016.2 cabinet; it offsets it in the other three 2014 ideological inversions and the remaining cabinet inversions. The balance between components varies even when every case satisfies the same inversion criterion.

These comparisons locate national differentials in the decomposition without assigning causal responsibility. B_C is not pure malapportionment; it combines district seat weights, turnout and valid-vote differences, and coalition vote geography. Likewise, party contributions in 2014 and 2018 are ex post attributions in elections where seats were often allocated through joint electoral lists. A large positive or negative party or district entry does not establish that the unit caused an inversion.

Brazil's multiple sources of disproportionality do not undermine the aggregation argument, but they delimit its interpretation. The analysis does not show that district magnitude, malapportionment, turnout, electoral coalitions, federations, thresholds, or rounding individually produced an inversion. It shows that the realized party differentials generated by this institutional environment can accumulate across particular political sets and cross a discontinuous majority threshold. In a pure-PR benchmark the whole-number problem is sufficient for this possibility; in Brazil it is one potential source among several.

Institutional complexity therefore weakens a monocausal claim but strengthens the case for studying the coalition transformation itself. Party differentials that appear modest or politically inconsequential in isolation may reinforce one another inside one coalition, cancel inside another, and leave a third just below 257 seats. The relevant contribution is not a claim that Brazil reveals a new source of disproportionality. It is the demonstration that ordinary differentials from several sources acquire a distinct institutional consequence when parties are recombined into realized or constrained potential majorities.

This limited claim nevertheless places the Brazilian cases within the broader literature on paradoxes and tradeoffs under proportional representation. Party-level proportionality does not settle the majority status of every post-electoral party set. Coalition inversions identify the narrower circumstance in which the ordinary vote–seat translation, once aggregated across a politically relevant set, yields a legislative majority without a corresponding national vote majority (Van

Deemen 1993; Kurrild-Klitgaard 2008; Kaminski 2018; Flis, Grofman, and Kaminski 2025).

Two further qualifications remain. The one-gap relaxation weakens exact connectedness while holding the estimated party order fixed. It therefore addresses whether one omitted intermediate party determines admissibility, but it does not incorporate uncertainty in the ideology estimates themselves; alternative placements may change endpoints, spans, or marginal cases. In addition, the vote share of a cabinet or potential coalition is the sum of votes cast for parties later grouped into that set, not a direct vote for the coalition. The criterion measures coalitional representation; it does not measure voter endorsement, legislative discipline, roll-call support, or portfolio bargaining. The analysis identifies a representational condition, not its public reception or normative legitimacy.

6 Conclusion

This article extends the literature on coalition inversions by following cabinet composition across legislative terms, comparing those realized party sets with ideologically constrained potential coalitions, and examining both in a multidistrict proportional system without national compensation. Party-level deviations can aggregate into politically consequential coalition differentials, but whether they change majority status depends on coalition configuration.

The empirical comparison is not uniform across elections or domains. Four observed cabinet periods invert: two linked to the 2014 election, one 238-day PSC-inclusive period linked to 2018, and the initial Lula cabinet linked to 2022. Inversions also occur among minimal connected winning coalitions in all three elections. PP–PL provides a compact seven-party right-side inversion in 2022, while PT–PSDB has a narrow vote minority in 2018. The one-gap relaxation admits more minimal inversions in every election, preserving the strongest 2014 and 2022 member sets and producing a stronger 2018 case.

The coalition representation ratio clarifies the threshold logic. Most observed cabinet coalitions are overrepresented, but only four combine that advantage with a vote minority and a seat majority. R_C measures relative advantage, d_C measures its absolute seat value, and q_C determines the coalition's proportional distance from 257. Coalition inversion is therefore not synonymous with overrepresentation. It is the subset of overrepresented configurations in which the accumulated seat surplus is majority-changing.

Brazil's multiple sources of disproportionality do not undermine the aggregation argument, but they delimit its interpretation. The analysis does not show that district magnitude, malapportionment, turnout, electoral coalitions, federations, thresholds, or rounding individually produced an inversion. It shows that the realized party differentials generated by this institutional environment can accumulate across particular political sets and cross a discontinuous majority threshold. In a pure-PR benchmark the whole-number problem is sufficient for this possibility; in Brazil it is one potential source among several.

Institutional complexity therefore weakens a monocausal claim but strengthens the case for studying the coalition transformation itself. Party differentials that appear modest or politically inconsequential in isolation may reinforce one another inside one coalition, cancel inside another, and leave a third just below 257 seats. The relevant contribution is not a claim that Brazil reveals a new source of disproportionality. It is the demonstration that ordinary differentials from several sources acquire a distinct institutional consequence when parties are recombined into realized or constrained potential majorities.

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Data availability. For peer review, the data and replication materials are available in an anonymized repository: https://anonymous.4open.science/r/electoral-inversion-B6B5/README.md. For final publication, the analysis uses public election data from Brazil's Tribunal Superior Eleitoral and cabinet/party classification inputs documented in the replication repository.

Code availability. For peer review, the code necessary to reproduce the analysis is available in an anonymized repository: https://anonymous.4open.science/r/electoral-inversion-B6B5/README.md. For final publication, the replication code is available at the public repository after acceptance.

Ethics approval. This study uses publicly available aggregate electoral, cabinet, and party-classification data. No new human participants were recruited by the author.

Consent to participate. Not applicable. This study uses publicly available aggregate and documentary data.

Consent for publication. Not applicable. This study uses publicly available aggregate and documentary data.

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A Seat-differential diagnostics

Party contributions identify which coalition members add to or offset d_C. The A_C and B_C terms partition the coalition differential into within-district allocation and between-district weighting, while district electoral weights provide a separate institutional diagnostic. These quantities are descriptive and do not identify causal effects.

A.1 Party contributions to coalition differentials

Table 4 reports the member-party composition of d_C for the focal cabinet and primary minimal parliamentary inversions. The complete machine-readable output includes all four cabinet inversions and all twelve primary exact-connected $k = 0$ ideological inversions, including the seven minimal cases, together with the all-party sensitivity results.

Table 4: Party contributions to coalition differentials

Case	d_C	Gross +	Gross -	Two largest positive d_i	Two largest negative d_i
Cabinet 2014/2016.2	20.271	22.773	2.502	PMDB (8.13); PSD (4.55)	PT (-2.42); PCdoB (-0.08)
Cabinet 2014/2017.1	9.952	17.778	7.826	PMDB (8.13); PSD (4.55)	PSDB (-4.43); PV (-2.56)
Cabinet 2018/2021.3/2022.1	14.624	27.536	12.912	PP (8.76); PR (5.72)	PSL (-7.70); PATRIOTA (-2.48)
Cabinet 2022/2023.1	12.204	26.461	14.257	UNIÃO (11.10); PT (6.94)	PSOL (-6.06); PSB (-5.57)
Ideological 2014/PSB-PTN	9.032	19.612	10.580	PMDB (8.13); PSD (4.55)	PSDB (-4.43); PT DO B (-3.28)
Ideological 2014/PTB-PR	13.601	23.526	9.925	PMDB (8.13); PSD (4.55)	PSDB (-4.43); PT DO B (-3.28)
Ideological 2014/PT DO B-PSDC	6.442	19.369	12.927	PMDB (8.13); PSD (4.55)	PSDB (-4.43); PT DO B (-3.28)
Ideological 2014/SOLIDARIEDADE-PSL	6.462	19.369	12.907	PMDB (8.13); PSD (4.55)	PSDB (-4.43); PSL (-3.26)
Ideological 2018/PT-PSDB	10.751	24.401	13.650	MDB (5.60); PSD (4.99)	PV (-4.32); REDE (-3.26)
Ideological 2022/MDB-UNIÃO	12.507	29.495	16.988	UNIÃO (11.10); PP (6.24)	PTB (-5.67); PODE (-4.94)
Ideological 2022/PP-PL	25.813	35.441	9.628	PL (13.78); UNIÃO (11.10)	NOVO (-3.35); PATRIOTA (-3.16)

Notes: Gross - is the absolute sum of negative party contributions. The three aggregate columns use a closure-preserving three-decimal display; all identities are calculated and checked exactly before rounding. d_i is an ex post accounting contribution to the coalition differential. Positive and negative party contributions sum to d_C. For 2014 and 2018, party-level contributions are ex post accounting attributions because seats were often allocated through joint electoral lists. These values do not identify party-specific causal effects of the electoral rules.

A.2 District contributions to cabinet inversions

Table 5 summarizes the district contributions to A_C in the four distinct cabinet inversions. Positive and negative sums are calculated separately across districts. Concentration is measured relative to the positive sum, $\sum_d \max(A_{Cd}, 0)$, before subtracting negative contributions.

Table 5: District contributions to the within-district component of cabinet inversions

Election	Period	Districts positive/negative	Positive sum	Negative sum	Top three share (%)	Districts for 90%
2014	2016.2	20/7	20.42	-4.67	45.1	13
2014	2017.1	20/7	15.96	-4.69	34.1	14
2018	2021.3/2022.1	22/5	21.96	-2.89	26.3	15
2022	2023.1	18/9	20.84	-7.64	37.4	12

Notes: Sums are measured in seats. Concentration columns refer to positive district contributions only.

A.3 District electoral weights

The neutral district-weight diagnostic compares each district's seats per valid vote with the national rate:

$$w_d = \frac{S_d/V_d}{S/V} = \frac{S_d/S}{V_d/V}.$$

Figure 5 shows an interpretable but only moderate inverse relationship between district magnitude and electoral weight. São Paulo is the unique 70-seat, lowest-weight district in all three elections, whereas the eleven eight-seat districts span values below the national average to more than five times it. Magnitude alone therefore does not define a homogeneous weighting category, and the plot is not coalition-specific.

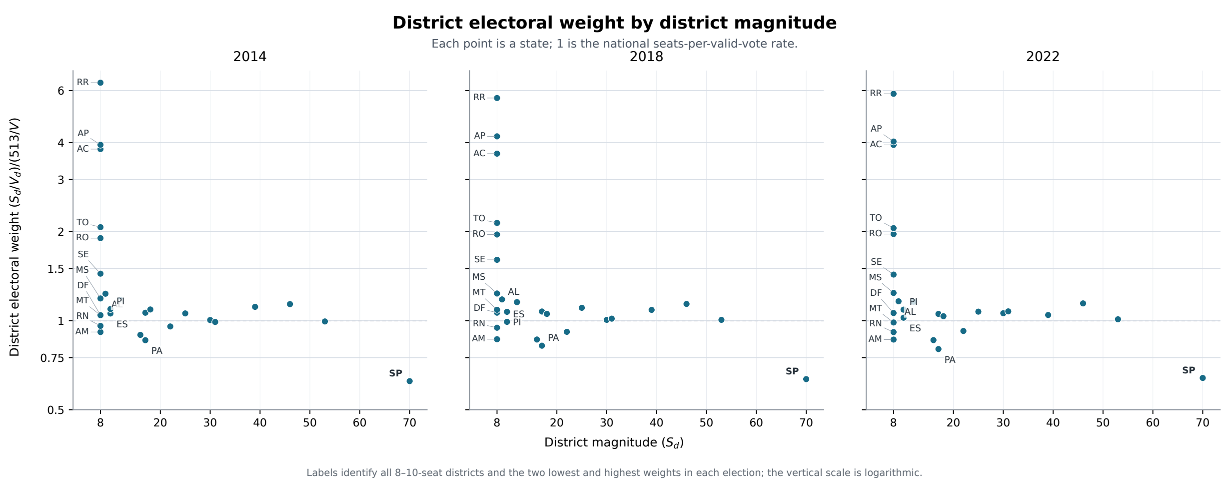

Figure 5: District electoral weight by district magnitude

Notes: Each point is a state or the Federal District. Electoral weight $w_d = 1$ is the national seats-per-valid-vote rate; the vertical scale is logarithmic. Labels identify all 8–10-seat districts and the two lowest and highest weights in each election. This is a measurement diagnostic, not an estimate of a district-magnitude effect or a coalition-specific contribution to B_C.

B Cabinet periods and composition

Table 6: Cabinet-period composition and transitions after cabinet-label harmonization and election-year label translation

Election	Period	Dates	Days	Vote %	Seats	Seat diff.	Status	Composition	Entered	Left
2014	2015.1	2015-01-01–2016-03-23	448	57.69	318	22.07	votes+seats	PCdoB, PDT, PMDB, PP, PR, PRB, PSD, PT, PTB		
2014	2016.1	2016-03-24–2016-04-13	21	53.14	297	24.39	votes+seats	PCdoB, PDT, PMDB, PP, PR, PSD, PT, PTB		PRB
2014	2016.2	2016-04-14–2016-04-15	2	46.54	259	20.27	seats only	PCdoB, PDT, PMDB, PR, PSD, PT, PTB		PP
2014	2016.3	2016-04-16–2016-05-11	26	40.41	223	15.72	neither	PCdoB, PDT, PMDB, PR, PT, PTB		PSD
2014	2016.4	2016-05-12–2017-12-27	595	53.93	291	14.33	votes+seats	DEM, PMDB, PP, PPS, PSB, PSD, PSDB, PTB, PV	DEM, PP, PPS, PSB, PSD, PSDB, PV	PCdoB, PDT, PR, PT
2014	2017.1	2017-12-28–2018-04-06	100	49.91	266	9.95	seats only	DEM, PMDB, PP, PPS, PSB, PSD, PSDB, PV		PTB
2014	2018.1	2018-04-07–2018-05-27	51	37.22	203	12.07	neither	PMDB, PP, PPS, PSD, PSDB		DEM, PSB, PV
2014	2018.2	2018-05-28–2018-12-31	218	37.96	207	12.26	neither	PMDB, PP, PPS, PSD, PSDB, PTN	PTN	
2018	2019.1	2019-01-01–2020-02-10	406	36.48	190	2.86	neither	DEM, MDB, NOVO, PATRIOTA, PR, PRB, PSL		
2018	2020.1	2020-02-11–2020-02-14	4	42.47	220	2.11	neither	DEM, MDB, NOVO, PATRIOTA, PR, PRB, PSDB, PSL	PSDB	
2018	2020.2	2020-02-15–2020-06-16	123	36.94	186	-3.50	neither	DEM, NOVO, PATRIOTA, PR, PRB, PSDB, PSL		MDB
2018	2020.3	2020-06-17–2020-12-08	175	42.79	221	1.49	neither	DEM, NOVO, PATRIOTA, PR, PRB, PSD, PSDB, PSL	PSD	
2018	2020.4	2020-12-09–2020-12-09	1	44.54	228	-0.49	neither	DEM, NOVO, PATRIOTA, PR, PRB, PSC, PSD, PSDB, PSL	PSC	
2018	2020.5	2020-12-10–2021-03-28	109	32.90	176	7.21	neither	DEM, NOVO, PATRIOTA, PR, PRB, PSC, PSD, PSDB		PSL
2018	2021.1	2021-03-29–2021-06-23	87	44.54	228	-0.49	neither	DEM, NOVO, PATRIOTA, PR, PRB, PSC, PSD, PSDB, PSL	PSL	
2018	2021.2	2021-06-24–2021-08-03	41	41.74	220	5.86	neither	DEM, PATRIOTA, PR, PRB, PSC, PSD, PSDB, PSL		NOVO
2018	2021.3/2022.1	2021-08-04–2022-03-29	238	47.25	257	14.62	seats only	DEM, PATRIOTA, PP, PR, PRB, PSC, PSD, PSDB, PSL	PP	
2018	2022.2	2022-03-30–2022-03-31	2	45.50	250	16.60	neither	DEM, PATRIOTA, PP, PR, PRB, PSD, PSDB, PSL		PSC
2018	2022.3	2022-04-01–2022-04-01	1	39.50	220	17.36	neither	DEM, PATRIOTA, PP, PR, PRB, PSD, PSL		PSDB
2018	2022.4	2022-04-02–2022-12-31	274	34.18	187	11.64	neither	DEM, PATRIOTA, PP, PRB, PSD, PSL		PR
2022	2023.1	2023-01-01–2023-09-12	255	48.89	263	12.20	seats only	MDB, PCdoB, PDT, PSB, PSD, PSOL, PT, REDE, UNIÃO		
2022	2023.2	2023-09-13–2025-12-23	833	63.79	350	22.76	votes+seats	MDB, PCdoB, PDT, PP, PSB, PSD, PSOL, PT, REDE, REPUBLICANOS, UNIÃO	PP, REPUBLICANOS	
2022	2025.1	2025-12-24–2026-03-19	86	54.45	291	11.66	votes+seats	MDB, PCdoB, PDT, PP, PSB, PSD, PSOL, PT, REDE, REPUBLICANOS		UNIÃO

Notes: Composition, Entered, and Left are reported using the election-year party labels used for vote-seat accounting. Period windows are constructed from the party labels observed in the cabinet data. When a party fusion occurs, the successor party is used to determine whether the cabinet party set changes; it is then expanded into antecedent election parties if the relevant election occurred before the fusion.

C Exact-connected minimal winning ideological intervals

Table 7 reports all minimal connected winning intervals in the primary $k = 0$ domain of Chamber-represented parties. These are contiguous intervals that reach the Chamber majority threshold and cease to do so when either endpoint is removed. The table is broader than Table 3, which reports only minimal connected interval inversions. The distinction matters because a minimal connected winning interval may have both a vote majority and a seat majority, while an interval inversion has a seat majority without a vote majority.

Table 7: Minimal connected winning ideological intervals

Election	Start	End	Parties	Vote %	Seats	Seat diff.	Status	Contains PT	Interval
2014	PCdoB	PMDB	12	50.17	265	7.63	votes+seats	Yes	PCdoB, PT, PDT, PSB, PPS, PV, PTB, PT DO B, SOLIDARIEDADE, PMN, PHS, PMDB
2014	PT	PSD	12	54.34	291	12.26	votes+seats	Yes	PT, PDT, PSB, PPS, PV, PTB, PT DO B, SOLIDARIEDADE, PMN, PHS, PMDB, PSD
2014	PDT	PSDB	12	51.80	276	10.25	votes+seats	No	PDT, PSB, PPS, PV, PTB, PT DO B, SOLIDARIEDADE, PMN, PHS, PMDB, PSD, PSDB
2014	PSB	PTN	12	48.92	260	9.03	seats only	No	PSB, PPS, PV, PTB, PT DO B, SOLIDARIEDADE, PMN, PHS, PMDB, PSD, PSDB, PTN
2014	PTB	PR	13	47.45	257	13.60	seats only	No	PTB, PT DO B, SOLIDARIEDADE, PMN, PHS, PMDB, PSD, PSDB, PTN, PRTB, PROS, PRP, PR
2014	PT DO B	PSDC	15	48.84	257	6.44	seats only	No	PT DO B, SOLIDARIEDADE, PMN, PHS, PMDB, PSD, PSDB, PTN, PRTB, PROS, PRP, PR, PRB, PTC, PSDC
2014	SOLIDARIEDADE	PSL	15	48.84	257	6.46	seats only	No	SOLIDARIEDADE, PMN, PHS, PMDB, PSD, PSDB, PTN, PRTB, PROS, PRP, PR, PRB, PTC, PSDC, PSL
2014	PMDB	PP	13	51.23	272	9.19	votes+seats	No	PMDB, PSD, PSDB, PTN, PRTB, PROS, PRP, PR, PRB, PTC, PSDC, PSL, PP
2018	PT	PSDB	14	49.95	267	10.75	seats only	Yes	PT, PDT, PSB, REDE, PPS, PV, PTB, AVANTE, SOLIDARIEDADE, PMN, PHS, MDB, PSD, PSDB
2018	PDT	PR	18	50.59	268	8.47	votes+seats	No	PDT, PSB, REDE, PPS, PV, PTB, AVANTE, SOLIDARIEDADE, PMN, PHS, MDB, PSD, PSDB, PODE, PPL, PROS, PRP, PR
2018	PSB	PRB	18	51.02	269	7.26	votes+seats	No	PSB, REDE, PPS, PV, PTB, AVANTE, SOLIDARIEDADE, PMN, PHS, MDB, PSD, PSDB, PODE, PPL, PROS, PRP, PR, PRB
2018	SOLIDARIEDADE	PSL	15	50.12	262	4.86	votes+seats	No	SOLIDARIEDADE, PMN, PHS, MDB, PSD, PSDB, PODE, PPL, PROS, PRP, PR, PRB, PTC, DC, PSL
2018	PMN	NOVO	15	50.93	257	-4.29	votes+seats	No	PMN, PHS, MDB, PSD, PSDB, PODE, PPL, PROS, PRP, PR, PRB, PTC, DC, PSL, NOVO
2018	MDB	PP	14	54.34	285	6.24	votes+seats	No	MDB, PSD, PSDB, PODE, PPL, PROS, PRP, PR, PRB, PTC, DC, PSL, NOVO, PP
2018	PSD	PSC	14	50.55	258	-1.34	votes+seats	No	PSD, PSDB, PODE, PPL, PROS, PRP, PR, PRB, PTC, DC, PSL, NOVO, PP, PSC
2018	PSDB	DEM	15	50.82	257	-3.72	votes+seats	No	PSDB, PODE, PPL, PROS, PRP, PR, PRB, PTC, DC, PSL, NOVO, PP, PSC, PATRIOTA, DEM
2022	PT	PP	15	57.18	284	-9.36	votes+seats	Yes	PT, PSB, REDE, PDT, PV, SOLIDARIEDADE, CIDADANIA, AVANTE, MDB, PSDB, PSD, PODE, PROS, PTB, PP
2022	PSB	PSC	16	53.82	261	-15.10	votes+seats	No	PSB, REDE, PDT, PV, SOLIDARIEDADE, CIDADANIA, AVANTE, MDB, PSDB, PSD, PODE, PROS, PTB, PP, REPUBLICANOS, PSC
2022	MDB	UNIÃO	10	49.22	265	12.51	seats only	No	MDB, PSDB, PSD, PODE, PROS, PTB, PP, REPUBLICANOS, PSC, UNIÃO
2022	PP	PL	7	45.26	258	25.81	seats only	No	PP, REPUBLICANOS, PSC, UNIÃO, PATRIOTA, NOVO, PL

D Sensitivity to the ideological party universe

The primary universe contains only parties winning Chamber seats, while the all-party sensitivity universe restores parties receiving valid votes but no seats to the ideological ordering. Both use all valid federal-deputy votes in the national denominator. Table 8 compares the number of parties, minimal winning coalitions, minimal inversions, and strongest cases for each election and admissibility condition. Complete coalition compositions, span and gap information, electoral metrics, and computed decomposition terms for both universes are supplied in the machine-readable replication outputs.

Table 8: Sensitivity to the ideological party universe

Election/k	Universe	$\lvert P^*\rvert$	MW	Inv.	Strongest inversion	Vote %	Seats
2014/0	Seat-winning	28	8	4	PTB–PR	47.45	257
2014/0	All parties	32	8	4	PTB–PR	47.59	257
2014/1	Seat-winning	28	87	46	PTB–PR	47.45	257
2014/1	All parties	32	88	43	PTB–PR, omitting PPL	47.45	257
2018/0	Seat-winning	30	8	1	PT–PSDB	49.95	267
2018/0	All parties	35	8	0	None	–	–
2018/1	Seat-winning	30	108	42	PCdoB–PODE, omitting PSDB	47.59	257
2018/1	All parties	35	118	24	PCdoB–PODE, omitting PSDB	47.82	257
2022/0	Seat-winning	23	4	2	PP–PL	45.26	258
2022/0	All parties	32	4	2	PP–PL	45.35	258
2022/1	Seat-winning	23	39	12	PP–PL	45.26	258
2022/1	All parties	32	53	17	PP–PL, omitting DC	45.26	258

Notes: $|P^*|$ is the number of parties in the specified ideological order; MW counts domain-minimal seat-majority coalitions, and Inv. counts inversions among them. Both universes use all valid federal-deputy votes in V. Strongest means the smallest national vote share, with ties resolved by membership count and coalition identifier. The strongest endpoint region changes for election/k = 2018/0; other comparisons preserve it.

The primary orders contain 28, 30, and 23 parties in 2014, 2018, and 2022; the all-party orders contain 32, 35, and 32. Exact-connected minimal-majority counts are unchanged, but the all-party universe has six minimal inversions instead of seven: its 2018 PT–PSDB interval includes the seatless PMB and has a vote majority. Under $k = 1$, the all-party inversion counts are 43, 24, and 17, compared with 46, 42, and 12 in the primary universe. The direction of the count difference therefore varies across elections.

The strongest 2014 and 2022 spans are PTB–PR and PP–PL in both universes. Their primary connected member sets coincide with the strongest all-party one-gap coalitions, which omit PPL and DC. In 2018, both one-gap searches select PCdoB–PODE omitting PSDB, but the primary coalition also excludes PMB by construction and receives 47.59 rather than 47.82 percent of all valid votes, with the same 257 seats. No vote denominator changes in these comparisons. Restoring extra-parliamentary parties changes which party sets are adjacent or admissible and which votes belong to coalition numerators.

The decomposition is sensitive in the 2022 MDB–UNIÃO case. Its primary within-district component is positive (2.95 seats), while the all-party component is negative (-1.29 seats); favorable between-district weighting supports inversion in both. The primary exact-connected inversions therefore all have positive within-district components, whereas five of the six all-party cases do. The existence of inversions and their strongest 2014 and 2022 locations survive, while the 2018 exact-connected result and this decomposition sign require the distinction between ideological universes.

Table 9 reports the all-party exact-connected minimal-inversion decomposition for direct comparison with Table 3.

Table 9: All-party sensitivity: exact-connected ($k = 0$) minimal inversions and seat-differential decomposition

Election	Start	End	Parties	Vote %	Seats	d_C	A_C	B_C
2014	PSB	PTN	12	48.92	260	9.032	10.509	-1.477
2014	PTB	PR	14	47.59	257	12.857	11.621	1.236
2014	PT DO B	PSDC	16	48.99	257	5.697	6.267	-0.570
2014	SOLIDARIEDADE	PSL	16	48.98	257	5.717	6.586	-0.869
2022	MDB	UNIÃO	15	49.98	265	8.591	-1.292	9.883
2022	PP	PL	8	45.35	258	25.355	22.821	2.534

E Cabinet coalitions and ideological intervals

All references to ideological intervals, connected closures, and minimal connected coalitions in this appendix section use the primary seat-winning universe and the exact-connected $k = 0$ baseline. The one-gap result is summarized separately in Table 2.

Table 10 compares observed cabinet coalitions with the ideological intervals that contain or approximate them. The bridge table keeps two operations separate: cabinet periods are defined from the party labels observed in the cabinet data, while ideological closures and vote-seat calculations use the translated election-year party set. The interval analysis therefore remains tied to the Chamber delegation produced by each election. The connected closure of a cabinet is the smallest no-gap interval in the filtered parliamentary order containing its represented election-year parties. Gaps count represented parties inside that closure that are not cabinet members. Cabinet votes and seats continue to sum the full translated cabinet party set; the restriction applies to the ideological diagnostics. The final columns report the closest minimal connected winning interval and, where one exists, the closest minimal connected inversion. This table links the observed coalition analysis to the interval analysis by showing whether realized cabinet coalitions are compact ideological blocs or sparse selections from a wider ideological span.

Table 10: Cabinet coalitions and ideological intervals

Election	Period	Cabinet	Ideology summary	Span	Gaps	Closure summary	Minimal winning	Minimal inversion
2014	2015.1	votes+seats; 57.69%; 318 seats	left: 2; center-left: 1; center-right: 1; right: 5	PCdoB–PP	15	left: 2; center-left: 2; center: 2; center-right: 5; right: 13	PCdoB–PMDB; 50.17%; 265 seats; overlap 5; J=0.31	PTB–PR; 47.45%; 257 seats; overlap 4; J=0.22
2014	2016.1	votes+seats; 53.14%; 297 seats	left: 2; center-left: 1; center-right: 1; right: 4	PCdoB–PP	16	left: 2; center-left: 2; center: 2; center-right: 5; right: 13	PCdoB–PMDB; 50.17%; 265 seats; overlap 5; J=0.33	PTB–PR; 47.45%; 257 seats; overlap 4; J=0.24
2014	2016.2	seats only; 46.54%; 259 seats	left: 2; center-left: 1; center-right: 1; right: 3	PCdoB–PR	12	left: 2; center-left: 2; center: 2; center-right: 5; right: 8	PCdoB–PMDB; 50.17%; 265 seats; overlap 5; J=0.36	PTB–PR; 47.45%; 257 seats; overlap 4; J=0.25
2014	2016.3	neither; 40.41%; 223 seats	left: 2; center-left: 1; center-right: 1; right: 2	PCdoB–PR	13	left: 2; center-left: 2; center: 2; center-right: 5; right: 8	PCdoB–PMDB; 50.17%; 265 seats; overlap 5; J=0.38	PTB–PR; 47.45%; 257 seats; overlap 3; J=0.19
2014	2016.4	votes+seats; 53.93%; 291 seats	center-left: 1; center: 2; center-right: 1; right: 4; far right: 1	PSB–DEM	15	center-left: 1; center: 2; center-right: 5; right: 14; far right: 2	PSB–PTN; 48.92%; 260 seats; overlap 7; J=0.50	PSB–PTN; 48.92%; 260 seats; overlap 7; J=0.50
2014	2017.1	seats only; 49.91%; 266 seats	center-left: 1; center: 2; right: 4; far right: 1	PSB–DEM	16	center-left: 1; center: 2; center-right: 5; right: 14; far right: 2	PSB–PTN; 48.92%; 260 seats; overlap 6; J=0.43	PSB–PTN; 48.92%; 260 seats; overlap 6; J=0.43
2014	2018.1	neither; 37.22%; 203 seats	center: 1; right: 4	PPS–PP	15	center: 2; center-right: 5; right: 13	PSB–PTN; 48.92%; 260 seats; overlap 4; J=0.31	PSB–PTN; 48.92%; 260 seats; overlap 4; J=0.31
2014	2018.2	neither; 37.96%; 207 seats	center: 1; right: 5	PPS–PP	14	center: 2; center-right: 5; right: 13	PSB–PTN; 48.92%; 260 seats; overlap 5; J=0.38	PSB–PTN; 48.92%; 260 seats; overlap 5; J=0.38
2018	2019.1	neither; 36.48%; 190 seats	right: 5; far right: 2	MDB–DEM	10	right: 15; far right: 2	PSDB–DEM; 50.82%; 257 seats; overlap 6; J=0.38	PT–PSDB; 49.95%; 267 seats; overlap 1; J=0.05
2018	2020.1	neither; 42.47%; 220 seats	right: 6; far right: 2	MDB–DEM	9	right: 15; far right: 2	PSDB–DEM; 50.82%; 257 seats; overlap 7; J=0.44	PT–PSDB; 49.95%; 267 seats; overlap 2; J=0.10
2018	2020.2	neither; 36.94%; 186 seats	right: 5; far right: 2	PSDB–DEM	8	right: 13; far right: 2	PSDB–DEM; 50.82%; 257 seats; overlap 7; J=0.47	PT–PSDB; 49.95%; 267 seats; overlap 1; J=0.05
2018	2020.3	neither; 42.79%; 221 seats	right: 6; far right: 2	PSD–DEM	8	right: 14; far right: 2	PSDB–DEM; 50.82%; 257 seats; overlap 7; J=0.44	PT–PSDB; 49.95%; 267 seats; overlap 2; J=0.10
2018	2020.4	neither; 44.54%; 228 seats	right: 7; far right: 2	PSD–DEM	7	right: 14; far right: 2	PSDB–DEM; 50.82%; 257 seats; overlap 8; J=0.50	PT–PSDB; 49.95%; 267 seats; overlap 2; J=0.10
2018	2020.5	neither; 32.90%; 176 seats	right: 6; far right: 2	PSD–DEM	8	right: 14; far right: 2	PSDB–DEM; 50.82%; 257 seats; overlap 7; J=0.44	PT–PSDB; 49.95%; 267 seats; overlap 2; J=0.10
2018	2021.1	neither; 44.54%; 228 seats	right: 7; far right: 2	PSD–DEM	7	right: 14; far right: 2	PSDB–DEM; 50.82%; 257 seats; overlap 8; J=0.50	PT–PSDB; 49.95%; 267 seats; overlap 2; J=0.10
2018	2021.2	neither; 41.74%; 220 seats	right: 6; far right: 2	PSD–DEM	8	right: 14; far right: 2	PSDB–DEM; 50.82%; 257 seats; overlap 7; J=0.44	PT–PSDB; 49.95%; 267 seats; overlap 2; J=0.10
2018	2021.3/2022.1	seats only; 47.25%; 257 seats	right: 7; far right: 2	PSD–DEM	7	right: 14; far right: 2	PSDB–DEM; 50.82%; 257 seats; overlap 8; J=0.50	PT–PSDB; 49.95%; 267 seats; overlap 2; J=0.10
2018	2022.2	neither; 45.50%; 250 seats	right: 6; far right: 2	PSD–DEM	8	right: 14; far right: 2	PSDB–DEM; 50.82%; 257 seats; overlap 7; J=0.44	PT–PSDB; 49.95%; 267 seats; overlap 2; J=0.10
2018	2022.3	neither; 39.50%; 220 seats	right: 5; far right: 2	PSD–DEM	9	right: 14; far right: 2	PSDB–DEM; 50.82%; 257 seats; overlap 6; J=0.38	PT–PSDB; 49.95%; 267 seats; overlap 1; J=0.05
2018	2022.4	neither; 34.18%; 187 seats	right: 4; far right: 2	PSD–DEM	10	right: 14; far right: 2	PSDB–DEM; 50.82%; 257 seats; overlap 5; J=0.31	PT–PSDB; 49.95%; 267 seats; overlap 1; J=0.05

Election	Period	Cabinet	Ideology summary	Span	Gaps	Closure summary	Minimal winning	Minimal inversion
2022	2023.1	seats only; 48.89%; 263 seats	far left: 1; left: 2; center-left: 3; center-right: 1; right: 1; far right: 1	PSOL–UNIÃO	11	far left: 1; left: 2; center-left: 4; center-right: 5; right: 5; far right: 3	PT–PP; 57.18%; 284 seats; overlap 6; J=0.33	MDB–UNIÃO; 49.22%; 265 seats; overlap 3; J=0.19
2022	2023.2	votes+seats; 63.79%; 350 seats	far left: 1; left: 2; center-left: 3; center-right: 1; right: 2; far right: 2	PSOL–UNIÃO	9	far left: 1; left: 2; center-left: 4; center-right: 5; right: 5; far right: 3	PT–PP; 57.18%; 284 seats; overlap 7; J=0.37	MDB–UNIÃO; 49.22%; 265 seats; overlap 5; J=0.31
2022	2025.1	votes+seats; 54.45%; 291 seats	far left: 1; left: 2; center-left: 3; center-right: 1; right: 2; far right: 1	PSOL–REPUBLICANOS	8	far left: 1; left: 2; center-left: 4; center-right: 5; right: 5; far right: 1	PT–PP; 57.18%; 284 seats; overlap 7; J=0.39	MDB–UNIÃO; 49.22%; 265 seats; overlap 4; J=0.25

The first result is that no observed cabinet coalition is itself a contiguous ideological interval. Every cabinet period has gaps inside its connected closure. The number of omitted represented parties ranges from seven to sixteen. Brazilian cabinet coalitions are therefore selective combinations within wider parliamentary ideological spans, omitting many represented parties between their leftmost and rightmost members.

This matters for the interpretation of cabinet inversions. The observed cabinet inversions are sparse coalitions, but their connected closures are not inversions. All connected closures in Table 10 are vote-and-seat majorities. The 2016.2 cabinet, for example, has 46.54 percent of the vote and 259 seats, but its connected closure runs from PCdoB to PR and contains 19 represented parties. That closure has 77.46 percent of the vote and 408 seats. The inversion therefore does not come from a compact left-to-right ideological bloc. It comes from a selective cabinet coalition that retains a seat majority while omitting twelve represented parties inside its ideological span.

The same logic applies to the more durable Temer-period inversion. The 2017.1 cabinet contains DEM, PMDB, PP, PPS, PSB, PSD, PSDB, and PV. Its connected closure runs from PSB to DEM, contains 24 represented parties, and has sixteen gaps. The observed cabinet has 49.91 percent of the vote and 266 seats, while the parliamentary PSB–DEM closure has 78.27 percent of the vote and 409 seats. The cabinet inversion is therefore not a connected ideological majority. It is a sparse center-to-right cabinet coalition located near the connected inversion region identified in the interval analysis. Its closest minimal connected winning interval is PSB–PTN, which is itself an inversion with 48.92 percent of the vote and 260 seats.

The Dilma coalitions have a different ideological profile from the later Temer coalitions. They are also sparse, but they are left-anchored. The cabinet sets before Temer include PT, PCdoB, and PDT while extending to parties such as PMDB, PSD, PR, PP, and PTB. Their closures run from PCdoB to either PR or PP, and their gap structure contains many center, center-right, and right parties that are not cabinet members. The impeachment transition changes this structure. From 2016.3 to 2016.4, PCdoB, PDT, PT, and PR leave, while PSB, PPS, PV, PSD, PSDB, PP, and DEM enter. In the ideology classification, the parties leaving are concentrated on the left and center-left, with one right party, while the parties entering are concentrated in the center, right, and far right. The cabinet remains punctured, but its ideological anchor shifts rightward.

The ideology diagnostics in the table confirm this transition. On the project's ideology scale, the average unweighted cabinet ideology rises from the pre-Temer sequence to the post-impeachment Temer sequence. The transition from 2016.3 to 2016.4 alone increases the unweighted cabinet ideology mean by 1.53 points and the seat-weighted mean by 1.49 points. This is the substantive difference between the two sparse coalition types. The pre-Temer coalitions are left-anchored and broad. The Temer coalitions are center-to-right and broad. Both are non-contiguous, but they occupy different regions of the ideological order.

The Bolsonaro-period cabinets are more concentrated on the right and far right, and their connected closures are narrower than the 2014 and 2022 closures. The PSC-inclusive 2021.3/2022.1 cabinet has 47.25 percent of the vote and exactly 257 seats. Its parliamentary closure, PSD–DEM, contains sixteen parties and seven gaps, has 56.67 percent of the vote and 292 seats, and is a vote-and-seat majority. Its closest minimal connected winning interval is PSDB–DEM, with 50.82 percent of the vote and 257 seats. The closest minimal connected inversion is PT–PSDB, with 49.95 percent and 267 seats, but its Jaccard overlap with the cabinet is only 0.095. The exact-connected inversion is therefore located in a different ideological region from the sparse right-side cabinet inversion.

The 2023.1 Lula cabinet returns to a left-anchored structure, but it is not a compact left coalition. It contains PSOL, PCdoB, PT, PDT, PSB, REDE, MDB, PSD, and UNIÃO. Its parliamentary closure runs from PSOL to UNIÃO, contains 20 represented parties, and has eleven gaps. The observed cabinet is an inversion, with 48.89 percent of the vote and 263 seats, but its connected closure has 79.83 percent of the vote and 407 seats. The closest minimal connected winning interval is PT–PP, a broad vote-and-seat majority with 57.18 percent of the vote and 284 seats. The closest minimal connected inversion is MDB–UNIÃO, which lies to the right of the cabinet's left anchor.

This comparison is crucial for interpreting the 2022 result. The Lula cabinet is a realized inversion, but it is not the compact connected inversion revealed by the interval search. The compact connected inversion is PP–PL: a seven-party right/far-right parliamentary interval with 45.26 percent of the vote and 258 seats. The observed Lula coalition reaches a seat majority through a broad and sparse cross-ideological cabinet arrangement. The exact-connected interval analysis shows that the most compact vote-minority seat-majority bloc in the same Chamber is located on the right.

The two analyses therefore meet at a specific structural point. Cabinet coalitions identify realized governing party sets. Connected intervals identify the seat-vote geometry of ideological blocs. The cabinet-to-interval comparison shows that realized cabinets may draw on seat asymmetries without being connected blocs themselves. In 2014, the pre-Temer coalitions are sparse and left-anchored, while the Temer coalitions are sparse and shifted rightward. In 2018, the PSC-inclusive cabinet period reaches exactly 257 seats without a vote majority, while its connected closure has both vote and seat majorities. In 2022, the Lula cabinet is a broad realized inversion, while the compact connected inversion lies on the right. The gap between cabinet coalitions and connected intervals is therefore not a nuisance. It shows how Brazil's coalition presidentialism combines sparse cabinet construction with an ideological distribution in which seat-efficient connected blocs are often located elsewhere.